

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -6

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Recap of Previous Lecture



doubt ?

Topic

Page Replacement Policies

Topic

Belady's Anomaly



Topics to be Covered



Topic

Frame Allocation

Topic

Multilevel Paging



Topic : Question

[GATE-2016]



#Q. Consider a computer system with ten physical page frames. The system is provided with an access sequence $a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20}$ where each a_i is page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is

1

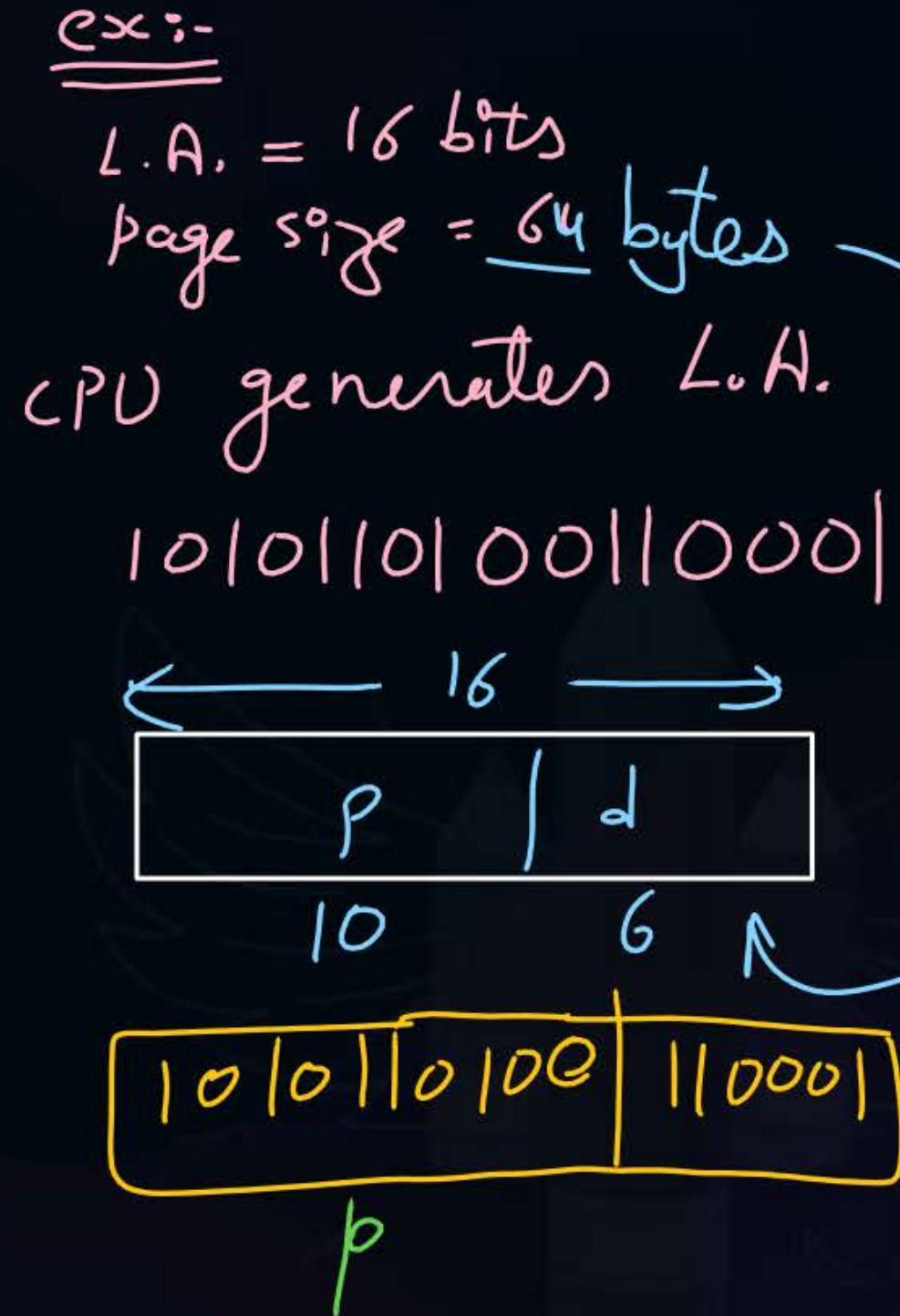
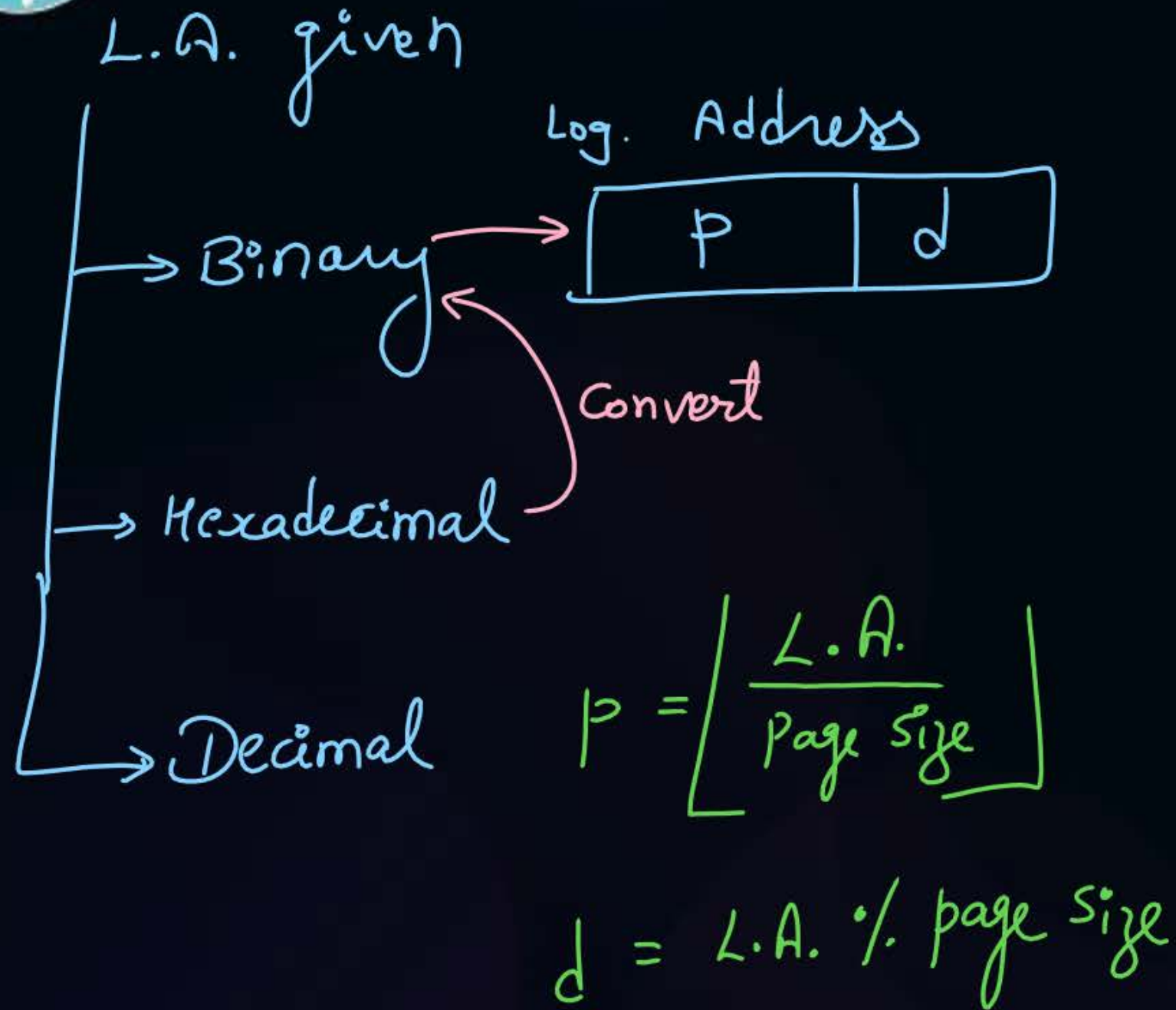
	P. f.
LIFO	31
optimal	30
Difference	1

$\left. \begin{matrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_9 \end{matrix} \right\}$ hits

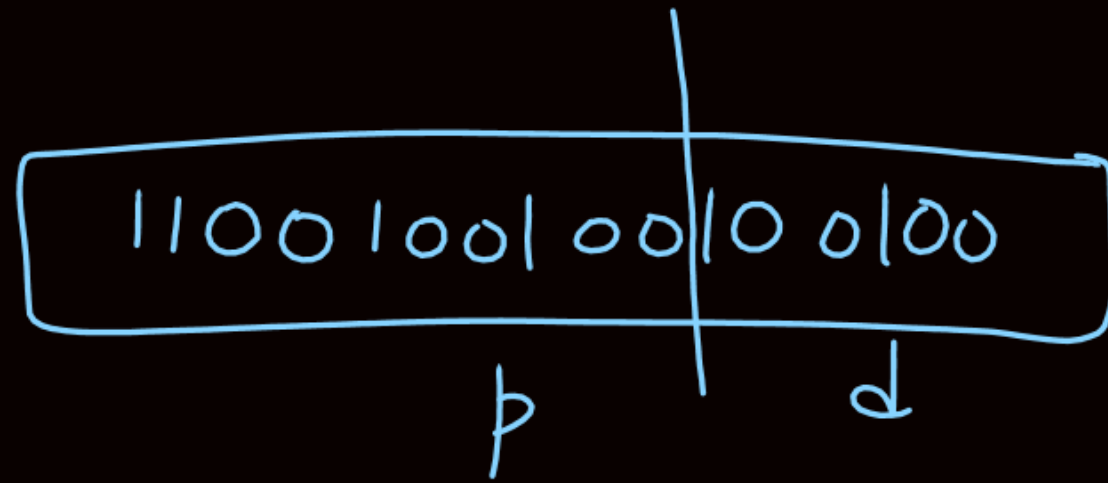
~~a_1~~ ~~a_2~~ ~~a_3~~ ~~a_4~~ ~~a_5~~ ~~a_6~~ ~~a_7~~ ~~a_8~~ ~~a_9~~ $\dots$ ~~a_{20}~~ ~~a_1~~ ~~a_2~~ $\dots$ a_{20}



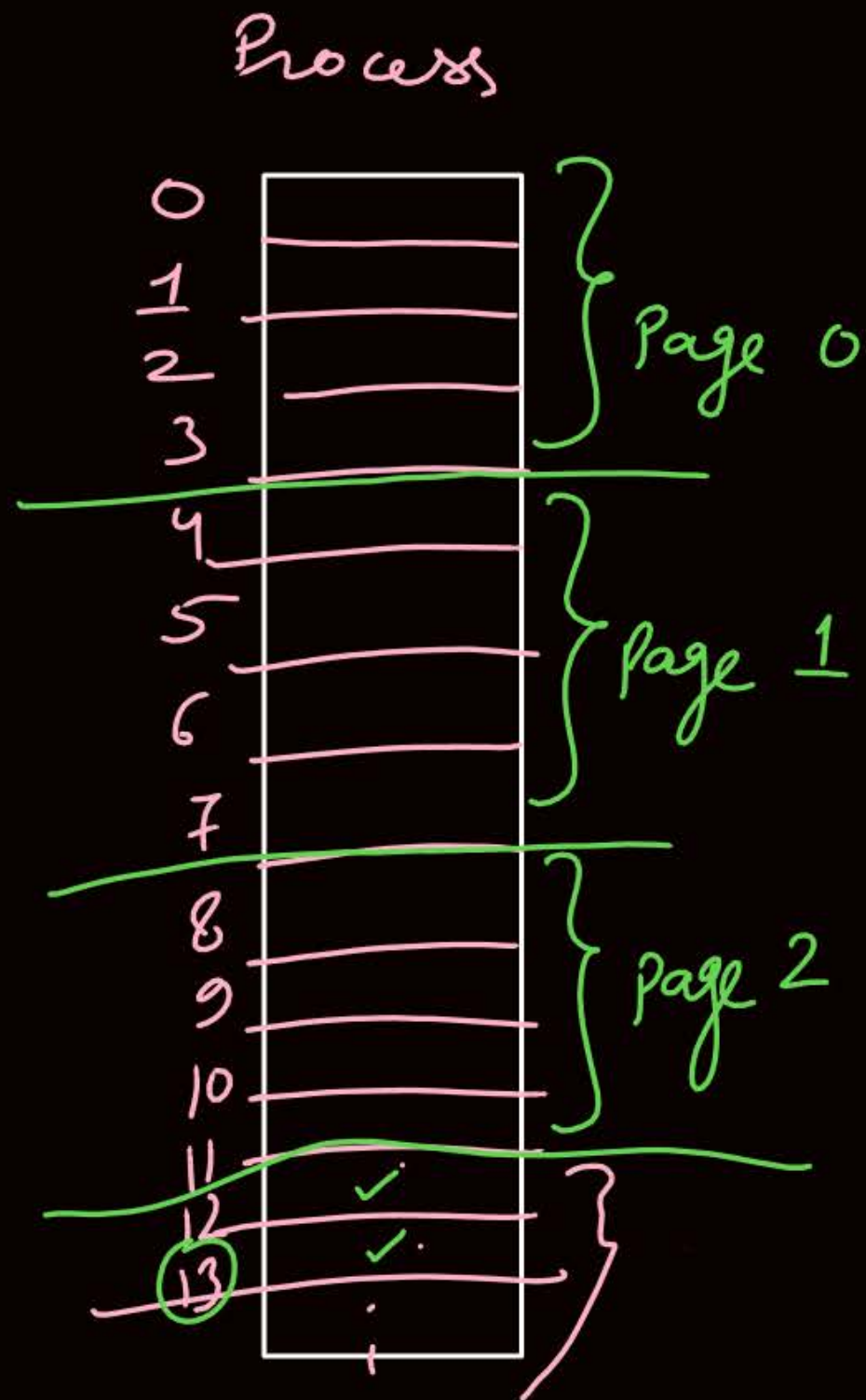
Topic : Making Page Reference Sequence



if L.A. $(C924)_{16}$



ex:-
assume page size = 4 bytes



$$L.A. = 4$$

$$p = \left\lfloor \frac{4}{4} \right\rfloor = 1$$

$$L.A. = 10$$

$$p = \left\lfloor \frac{10}{4} \right\rfloor = 2$$

$$L.A. = 11$$

$$p = \left\lfloor \frac{11}{4} \right\rfloor = 2$$

$$d = 11 \% 4 = 3$$

$$p = 3$$

$$d = 1$$

$$(3 \times 4) + 1 = 13$$

$$f = \left\lfloor \frac{P.A.}{\text{Page size}} \right\rfloor$$

$$d = P.A. \% \text{ page size}$$

if p and d given

$$L.A. = (p * \text{page size}) + d$$

$$P.A. = (f * \text{page size}) + d$$

ex:- Given a page table

Process P.T.

0	6
1	9
2	7
3	5

Page size = 16 bytes

$$L.A. = (51)_{10}$$

$$P.A. = (83)_{10} ?$$

Solⁿ

$$P = \left\lfloor \frac{51}{16} \right\rfloor = 3$$

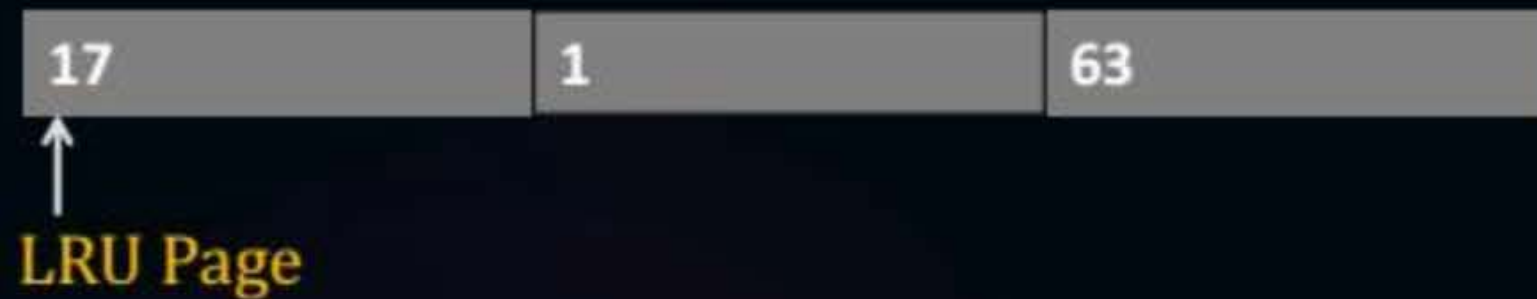
$$d = 51 \% 16 = 3$$

Search P.T. with $P=3$

$$f = 5$$

$$P.A. = (5 * 16) + 3 = \underline{\underline{83}} \text{ Ans.}$$

#Q. A demand paged virtual memory system uses 16 bit virtual address, page size of 256 bytes, and has 1Kbyte of main memory. LRU page replacement is implemented using the list, whose current status (page number is decimal) is



$$\text{no. of frames} = \frac{1\text{KB}}{256\text{B}} = 4$$

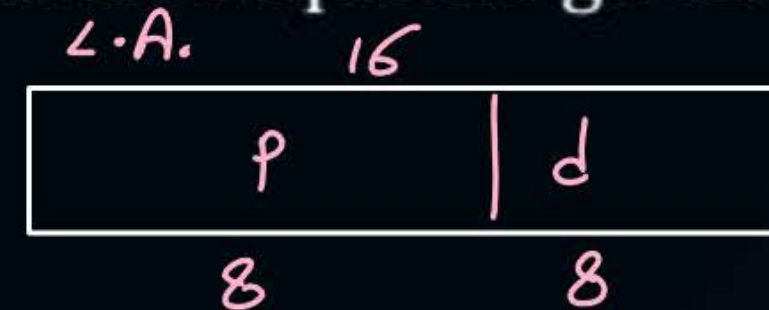
For each hexadecimal address in the address sequence given below,

00FF, 010D, 10FF, 11B0

indicate

1. the new status of the list
2. page faults, if any, and
3. page replacements, if any.

↓
2 times



$$1. (00000000)_2 = (0)_{10}$$

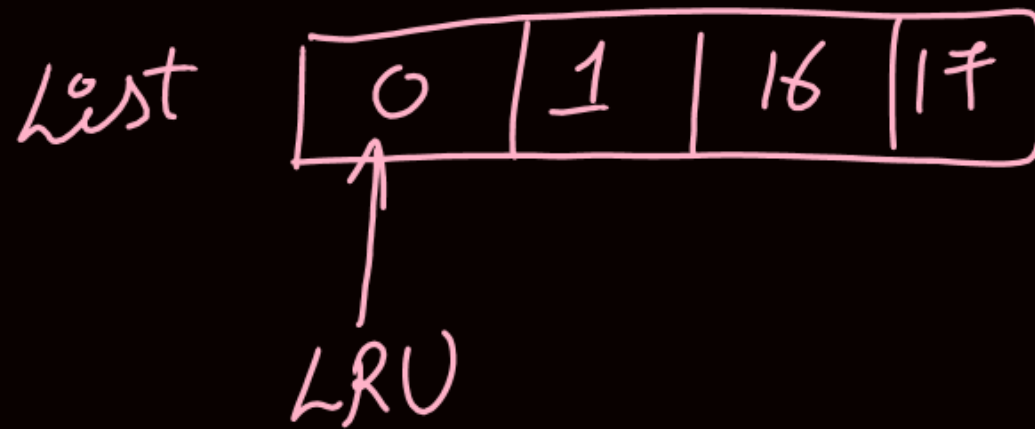
$$2. (00000001)_2 = (1)_{10}$$

$$3. (00010000)_2 = (16)_{10}$$

$$4. (00010001)_2 = (17)_{10}$$

Page reference seq :- $\overset{\checkmark}{0}, \overset{\text{hit}}{\overset{\checkmark}{1}}, \overset{\checkmark}{16}, \overset{\checkmark}{17}$

17	17	16	16
<u>1</u>	1	1	1
63	63	63	17
	0	0	0



#Q. Assume that a main memory with only 4 pages, each of 16 bytes, is initially empty. The CPU generates the following sequence of virtual addresses and uses the Least Recently Used (LRU) page replacement policy.

0, 4, 8, 20, 24, 36, 44, 12, 68, 72, 80, 84, 28, 32, 88, 92

How many page faults does this sequence cause? What are the page numbers of the pages present in the main memory at the end of the sequence?

✓ 0, ✓ 1, ✓ 2, ✗ 0, ✓ 4, ✓ 5, ✓ 1, ✓ 2, ✗ 5

A 6 and 1,2,3,4

B ✓ 7 and 1,2,4,5

C 8 and 1,2,4,5

D 9 and 1,2,3,5

0	0	0	0	0	0	2
	1	1	1	5	5	5
			2	2	1	1
		2	4	4	4	4



Topic : Frame Allocation



- How many frames do we allocate per process?





Topic : Frame Allocation

How many frames do we allocate per process?

- If it is a single-user, single-tasking system, it's simple – all the frames belong to the user's process





Topic : Frame Allocation

1. Equal Allocation $\rightarrow$ each process gets equal frames
2. Proportional Allocation $\rightarrow$ each process gets frames based on its size

ex:-

Total frames = 6

2 processes:-

P1 $\Rightarrow$ 6 pages

P2 $\Rightarrow$ 3 pages

equal allocation

P1 $\Rightarrow$ 3 frames

P2 $\Rightarrow$ 3 frames

Proportional allocation

P1 $\Rightarrow \frac{6}{9} * 6 = 4$ frames

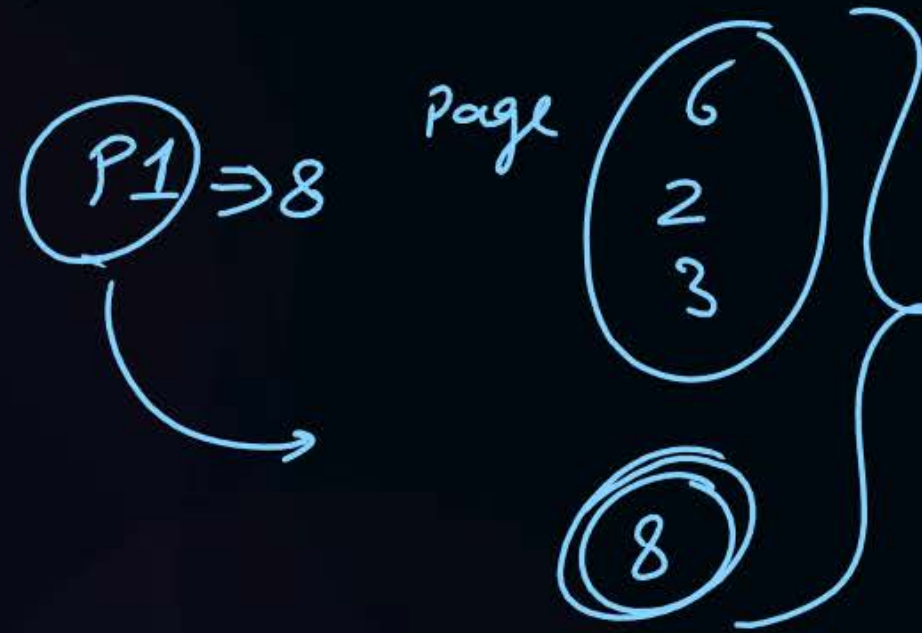
P2 $\Rightarrow \frac{3}{9} * 6 = 2$ frames



Topic : Frame Allocation

2 Questions

- What is the minimum number of frames that a process needs?
- Is page replacement global or local?





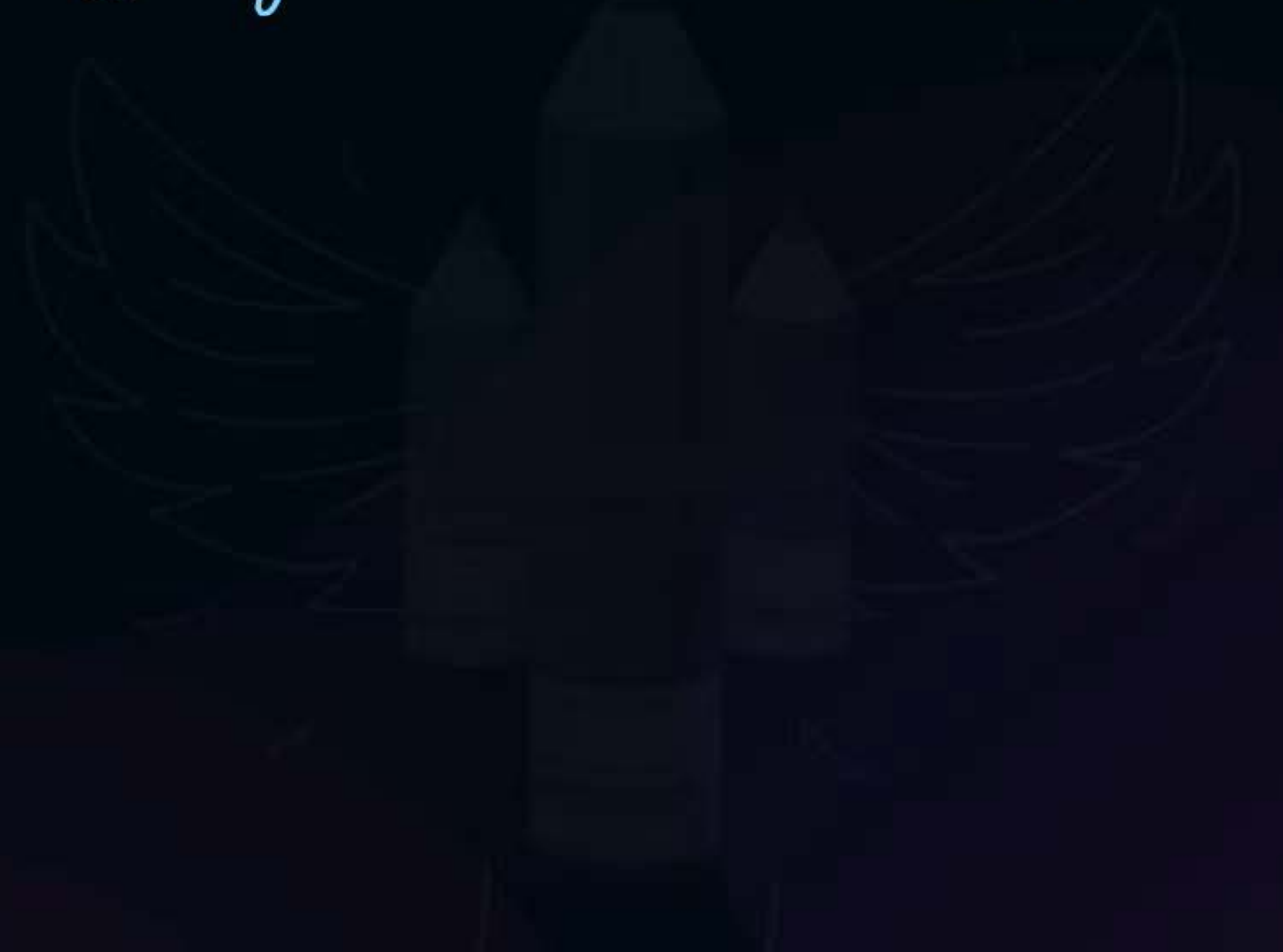
Topic : Minimum Number of Frames



Every process must have enough pages to complete an instruction.



$\text{min. page size} = \text{size of one instruction}$





Topic : Local Allocation

1. Local replacement requires that the page being replaced be in a frame belonging to the same process
2. The number of frames belonging to the process will not change
3. This allows processes to control their own page fault rate

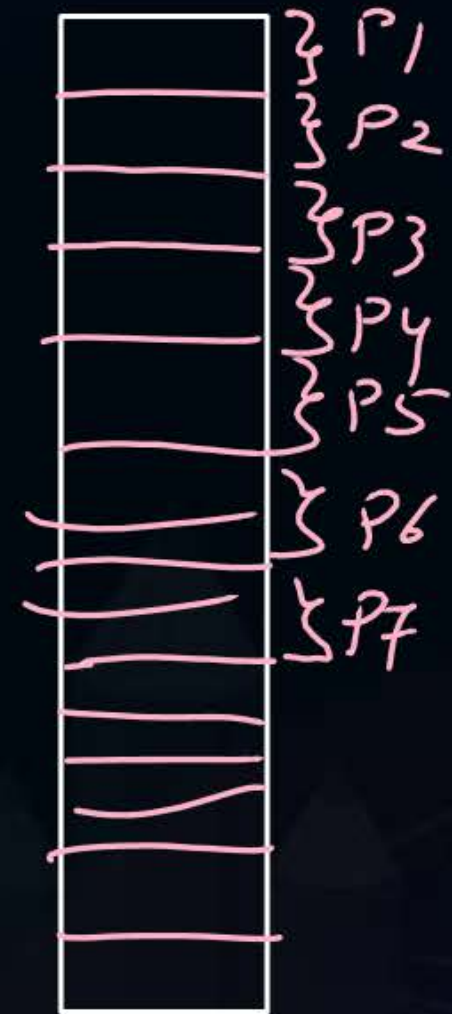
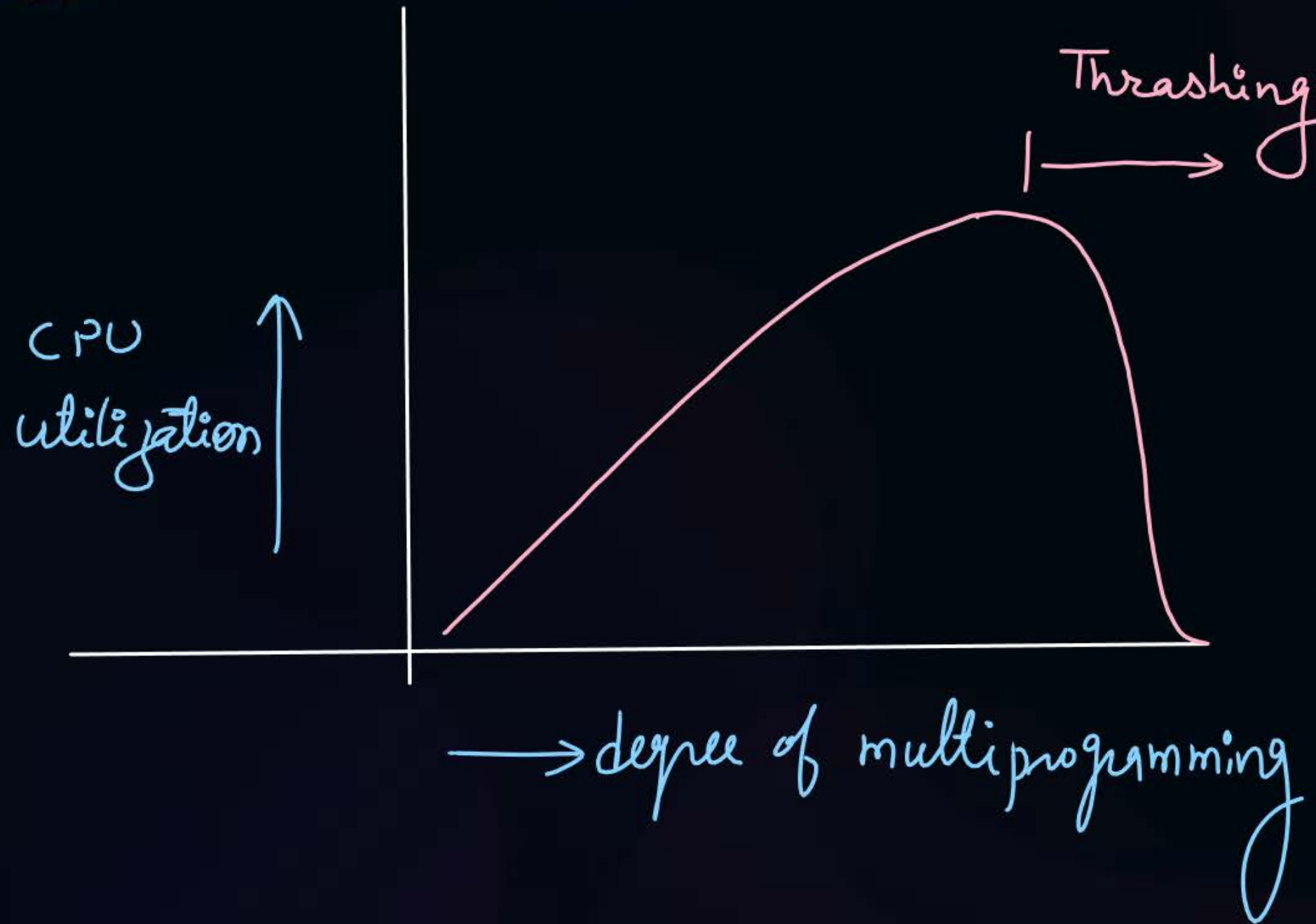


Topic : Global Allocation

1. The process can replace a page from a set that includes all the frames allocated to user processes
2. High-priority processes can increase their allocation at the expense of lower-priority processes
3. Global allocation makes for more efficient use of frames and their better throughput



Topic : Thrashing



Accessing page faults $\Rightarrow$ Thrashing (High-Level paging activity)

$\rightarrow$ CPU spends max or whole time only for p.f. service.



Topic : Locality Model

1. Working Set Model
2. Page Fault Frequency



Topic : Working Set

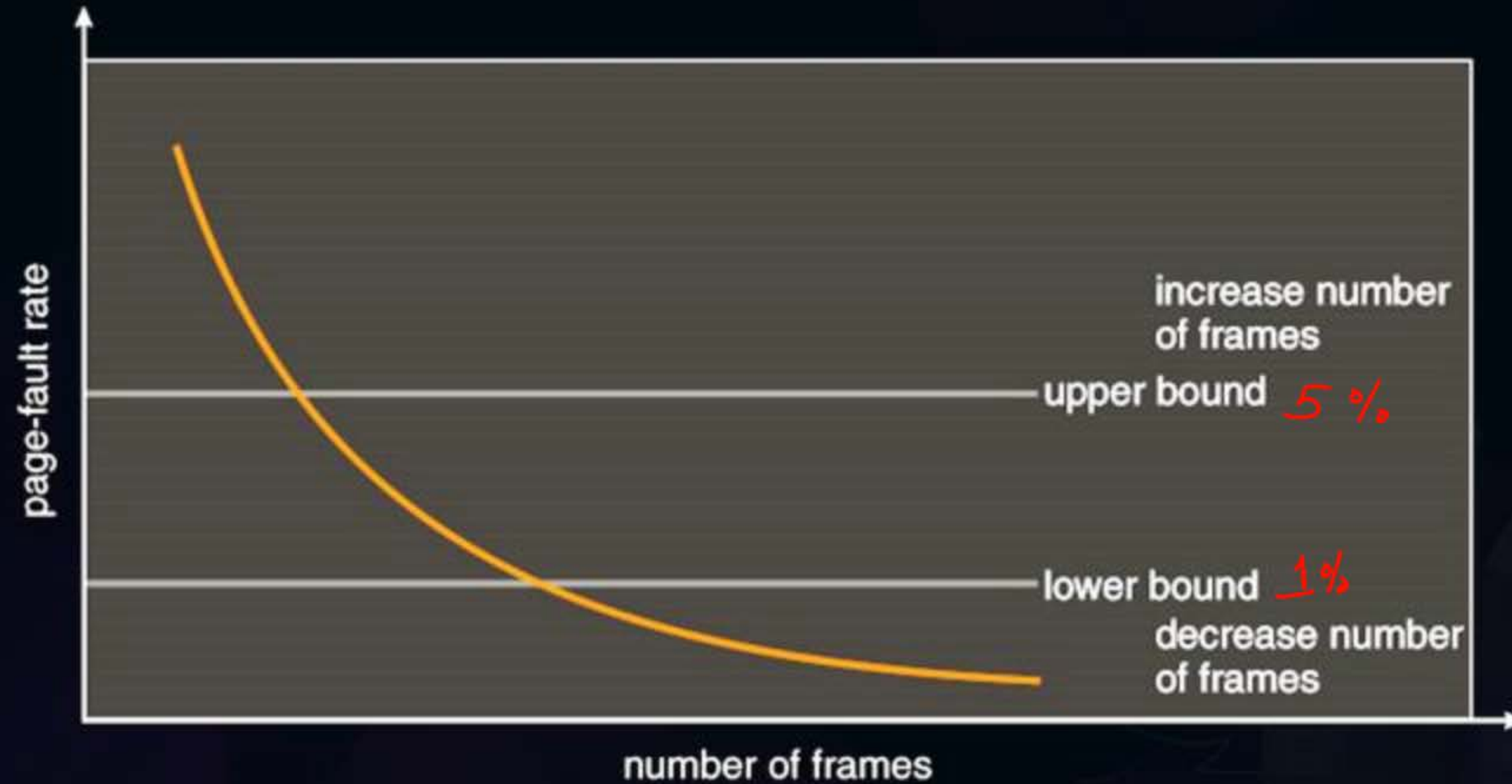


↓
OS predicts working set size of each process,
and tries to keep those many pages for
that process execution.





Topic : Page Fault Frequency

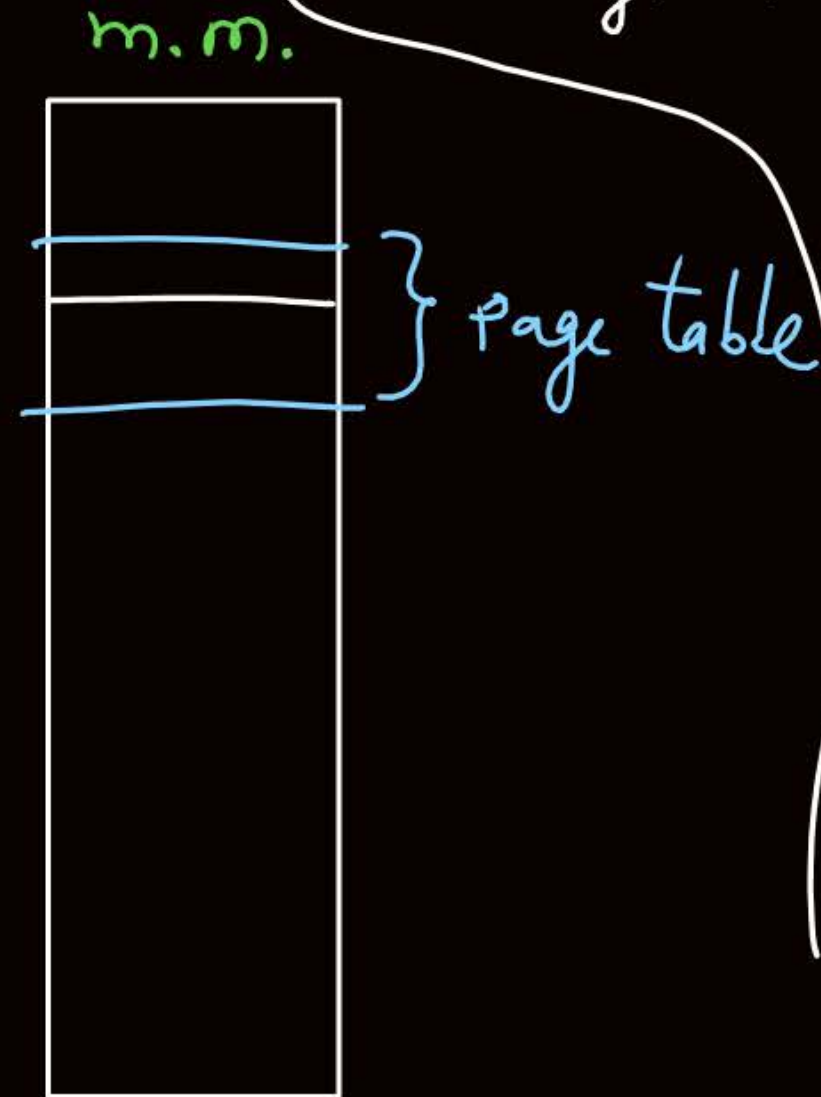
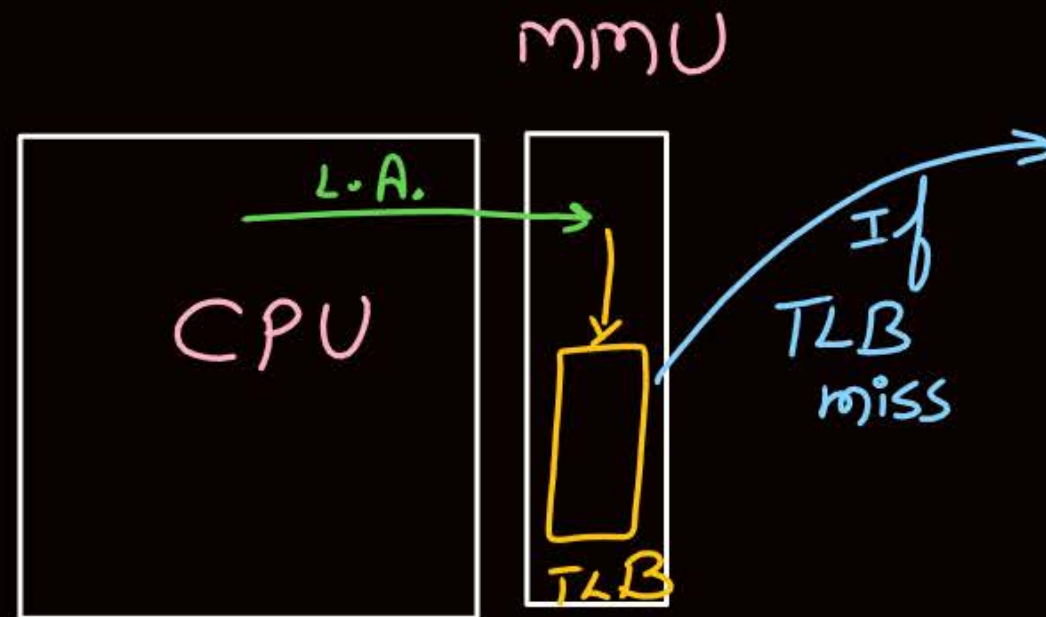


m.m.u. (Memory Management Unit) :- → It does L.A. to P.A. translation

↳ Hardware

→ Protection check

→ Page table updation during page fault service



→ change of valid, modified, replacement bits in P.T. entry



Topic : Let's Take a Simple Example

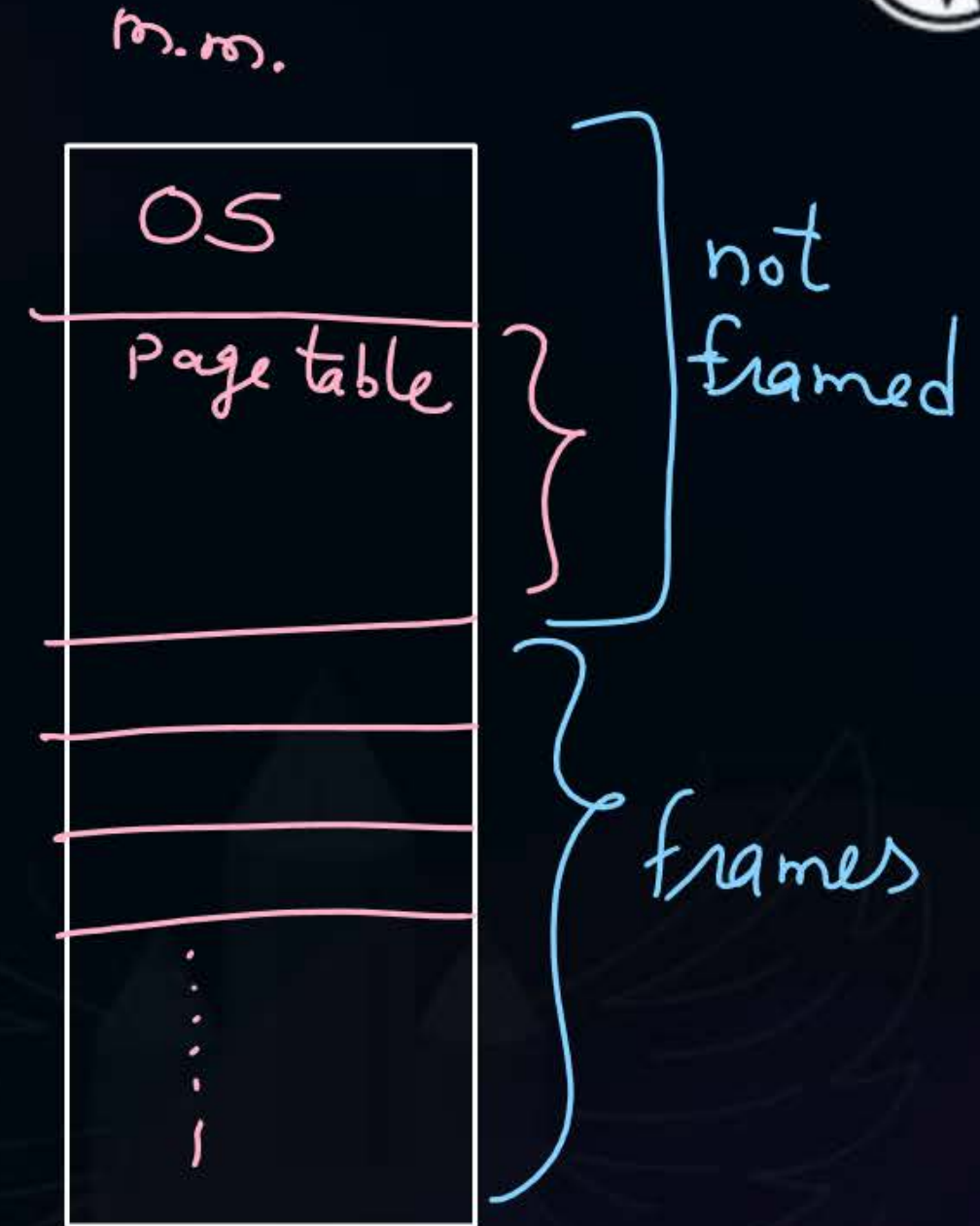
Process Size = 32B

Page size = 4B

Page table entry size = 1B

Page Table Size = ? $8 * 1B = 8B$

$$\left. \begin{array}{l} \text{Process Size} = 32B \\ \text{Page size} = 4B \end{array} \right\} \text{no. of pages} = \frac{32B}{4B} = 8$$





Topic : Now What?



Can you keep entire page table in single page?

If not then how to access the specific entry?



if page table also should be stored
on frames in form of pages then
multilevel paging is used.



2 mins Summary

Topic

Frame Allocation

Topic

Multilevel Paging



Happy Learning

THANK - YOU

